ENVIRONMENTAL ENGINEERING

http://www.ccee.iastate.edu/

Administered by the Department of Civil, Construction and Environmental Engineering

For undergraduate curriculum in Environmental Engineering leading to the degree Bachelor of Science. The Environmental Engineering Program is accredited by the Engineering Accreditation Commission of ABET, https://www.abet.org (https://www.abet.org/), under the commission's General Criteria and Program Criteria for Environmental Engineering and Similarly Named Engineering Programs.

Students in the environmental engineering bachelor's degree program will complete a curriculum covering the engineering and science knowledge necessary to design and implement effective, affordable solutions for environmental challenges involving water, air, and land. The environmental engineering curriculum equips students with a broad education that includes technical skills in analysis and design and professional practices such as communication, teamwork, leadership, and ethics. Graduates will have a strong foundation necessary to solve complex current and future infrastructure challenges within the diverse areas of environmental engineering.

Graduates of this program will be prepared to work in environmental engineering positions within the private and public (e.g., federal, military, state, and community) sectors that deal with pollution and contamination in all aspects of the built and natural environment. Examples of this work include analyzing and designing systems for water supply and distribution, collecting and processing waste, controlling air quality, recycling residuals, and protecting public health. Students interested in a more general education in civil engineering should consider the B.S. in civil engineering with environmental emphasis.

Student Learning Outcomes: Graduates of the Environmental Engineering curriculum should have, at the time of graduation:

1. An ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics.
2. An ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
3. An ability to communicate effectively with a range of audiences.
4. An ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts.
5. An ability to function effectively on a team whose members together provide leadership, create a collaborative environment, establish goals, plan tasks, and meet objectives.
6. An ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.
7. An ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Program Educational Objectives:

By three to five years after graduation, graduates of the civil, construction, and environmental engineering programs will have:

1. Pursued successful careers and expertise in civil engineering, construction engineering, environmental engineering, or a related profession that includes professionalism and increasing responsibility.
2. Collaborated effectively on multi-disciplinary teams to address the needs of society and the environment
3. Pursued lifelong learning, professional development, and licensure as appropriate for their career goals and for developments in the field.

The faculty encourages the students to develop their professional skills by participating in cooperative education, internships, or progressive summer engineering employment and study abroad programs. Qualified juniors and seniors interested in graduate studies may apply to the Graduate College to pursue concurrently the bachelor's degree and either a Master of Science in Civil Engineering or a Master of Business Administration in the College of Business Administration. These students would have an opportunity to graduate in five years with both degrees.

Curriculum in Environmental Engineering

Administered by the Department of Civil, Construction and Environmental Engineering.

Leading to the degree Bachelor of Science.

Total credits required: 131.

Any transfer credit courses applied to the degree program require a grade of C or better (but will not be calculated into the ISU cumulative GPA, Basic Program GPA or Core GPA). See also Basic Program and Special Programs. Note: Department does not allow Pass/Not Pass credits to be used to meet graduation requirements.

International Perspectives: 3 cr. [1]

U.S. Cultures and Communities: 3 cr. [1]

Communication Proficiency/Library requirement

ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3

ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course)	3
LIB 1600	Introduction to College Level Research	1

Social Sciences and Humanities: 12 cr. [2]

Complete 12 cr. with 6 cr. at 2000-level or above.

Basic Program: 24 cr [3]. Minimum GPA of 2.00 required for this set of courses to graduate, including any transfer courses (please note that transfer course grades will not be calculated into the Basic Program GPA).

CHEM 1770	General Chemistry I	4
or CHEM 1670	General Chemistry for Engineering Students	
ENGL 1500	Critical Thinking and Communication (Must have a C or better in this course)	3
ENGR 1010	Engineering Orientation	R
CE 1600	Systematic Problem Solving and Computer Applications for Civil, Construction, and Environmental Engineers [3]	3
LIB 1600	Introduction to College Level Research	1
MATH 1650	Calculus I	4
MATH 1660	Calculus II	4
PHYS 2310	Introduction to Classical Physics I	4
PHYS 2310L	Introduction to Classical Physics I Laboratory	1
Total Credits		**24**

Math and Physical Science: 27 cr.

CHEM 1770L	Laboratory in General Chemistry I	1
CHEM 1780	General Chemistry II [4]	3
CHEM 1780L	Laboratory in College Chemistry II [4]	1
CHEM 2310	Elementary Organic Chemistry	3
CHEM 2310L	Laboratory in Elementary Organic Chemistry	1
BIOL 2510	Biological Processes in the Environment	3
GEOL 2010	Geology for Engineers and Environmental Scientists	3
MATH 2650	Calculus III	4
MATH 2660	Elementary Differential Equations	3
MICRO 2010	Introduction to Microbiology	2
STAT 3005	Engineering Statistics	3-4
or STAT 3031	Probability and Statistical Inference for Engineers	
Total Credits		**27-28**

Env Engineering Core: 30 cr. Minimum GPA of 2.00 required for this set of courses to graduate (including transfer courses; please note that transfer course grades will not be calculated into the Core GPA).

ENVE 2010	Environmental Engineering Measurements and Analysis	3
ENVE 3260	Principles of Environmental Engineering	3
CE 3720	Engineering Hydrology and Hydraulics	3
ABE 3780	Mechanics of Fluids	3
ME 2310	Engineering Thermodynamics I	3
ENVE 4260	Environmental Engineering Science	3
ENVE 4270	Environmental Engineering Systems	3
ENVE 4280	Water and Wastewater Treatment Plant Design	3
ENVE 4290	Air Pollution and Control	3
ENVE 4300	Solid and Hazardous Waste Management	3
Total Credits		**30**

Other Remaining Courses: 38 cr.

ENGL 2500	Written, Oral, Visual, and Electronic Composition (Must have a C or better in this course)	3
SPCM 2120	Fundamentals of Public Speaking	3
ENVE 1200	Environmental Engineering Learning Community	1
ENVE 1900	Introduction to Undergraduate Research in Environmental Engineering	2
CE 3060	Project Management for Civil and Environmental Engineers	3
CE 2060	Engineering Economic Analysis and Professional Issues in Civil Engineering	3
CE 2710 & CE 2720	Engineering Foundations of Statics and Applied Engineering Statics	3
or CE 2740	Engineering Statics	
CE 3600	Geotechnical Engineering	4
EM 3240	Mechanics of Materials	3
ABE 3780L	Mechanics of Fluids Laboratory	1
Engineering Topics Electives [2]		6
Sustainability Elective [2]		3
Technical Communication Elective [2]		3
Total Credits		**38**

Seminar/Co-op/Internships: R cr.
Co-op/Internship optional.
Notes.

1. These university requirements will add to the minimum credits of the program unless the university-approved courses are also approved by the department to meet other course requirements within the degree program. U.S. Cultures and Communities and International Perspectives courses may not be taken Pass/Not Pass.
2. Choose from department approved list. (https://www.ccee.iastate.edu/undergraduate-programs/environmental-engineering/environmental-engineering-forms-and-information-by-cohort/)
3. See Basic Program for Professional Engineering Curricula for accepted substitutions for curriculum designated courses in the Basic Program.

First Year			
Fall	**Credits**	**Spring**	**Credits**
ENGR 1010	Required	ENVE 1900	2
ENVE 1200	1	MATH 1660	4
CE 1600	3	PHYS 2310	4
MATH 1650	4	PHYS 2310L	1
CHEM 1770	4	CE 2710	1
CHEM 1770L	1	ENGL 2500	3
ENGL 1500	3	SSH Elective	3
LIB 1600	1		
	17		**18**
Second Year			
Fall	**Credits**	**Spring**	**Credits**
ENVE 2010	3	ABE 3780	3
MATH 2650	4	ABE 3780L	1
CE 2720	2	BIOL 2510	3
CHEM 1780	3	CE 3060	3
CHEM 1780L	1	MATH 2660	3
SSH Elective	3	CHEM 2310	3
		CHEM 2310L	1
	16		**17**
Third Year			
Fall	**Credits**	**Spring**	**Credits**
ENVE 3260	3	CE 3720	3
MICRO 2010	2	ME 2310	3
EM 3240	3	GEOL 2010	3
CE 2060	3	CE 3600	4
STAT 3005	3	Technical Communication Elective	3
Sustainability Elective	3		
	17		**16**
Fourth Year			
Fall	**Credits**	**Spring**	**Credits**
ENVE 4260	3	ENVE 4270	3
ENVE 4300	3	ENVE 4280	3
Engineering Topics Elective	3	ENVE 4290	3
SPCM 2120	3	Engineering Topics Elective	3
International Perspective	3	US Cultures and Communities	3
	15		**15**